

OBSERVATIONS

Configuration	Waveform Appearance	Why it looks this way
Without LED	Ideal Pulsating DC. Smooth, rounded humps that touch (or nearly touch) the zero-line.	The circuit is linear. Only the bridge diodes (1.4V total) must be overcome, which happens very quickly.
LED After Bridge	Notched Humps. Sharp "V" shapes or spikes appear at the bottom of each hump.	The LED turns off abruptly when $V < V_f$. The "notch" is the measurement point reacting to the LED's non-linear resistance.
LED Before Bridge	Truncated Humps. Wide, flat horizontal lines (Dead Zones) appear between each hump.	The LED acts as a gate for the entire circuit. No current reaches the bridge or load until the AC peak exceeds $V_f + 1.4V$.